V2V-GoT: Vehicle-to-Vehicle Cooperative Autonomous Driving with Multimodal Large Language Models and Graph-of-Thoughts

***Abstract*— Current state-of-the-art autonomous vehicles could face safety-critical situations when their local sensors are occluded by large nearby objects on the road. Vehicle-to-vehicle (V2V) cooperative autonomous driving has been proposed as a means of addressing this problem, and one recently introduced framework for cooperative autonomous driving has further adopted an approach that incorporates a Multimodal Large Language Model (MLLM) to integrate cooperative perception and planning processes. However, despite the potential benefit of applying graph-of-thoughts reasoning to the MLLM, this idea has not been considered by previous cooperative autonomous driving research. In this paper, we propose a novel graph-of-thoughts framework specifically designed for MLLM-based cooperative autonomous driving. Our graph-of-thoughts includes our proposed novel ideas of occlusion-aware perception and planning-aware prediction. We curate the V2V-GoT-QA dataset and develop the V2V-GoT model for training and testing the cooperative driving graph-of-thoughts. Our experimental results show that our method outperforms other baselines in cooperative perception, prediction, and planning tasks. Our code and dataset are publicly released for open-source research.**

I. Introduction

Today's autonomous vehicles rely mainly on mounted cameras or LiDAR sensors to perceive the world , and some potential obstacles could be occluded by other large nearby objects, such as buses or trucks. To mitigate this safety issue, recent research has proposed V2V cooperative perception [1]–[5]. : each Connected Autonomous Vehicle (CAV) can share its individual perception information with others to improve overall detection and tracking accuracy.

One recent trend in autonomous driving research is using Large Language Models (LLMs) to build end-to-end driving systems [6]–[8] motivated by promising reasoning and generalization capabilities. However, most LLM-based driving research focuses on individual autonomous vehicles without cooperative perception or planning.

Some recent pioneering research works [9]–[12] have started to explore LLM-based cooperative autonomous driving. However, the V2V cooperation in prior works [10], [11] only involves the LLM as a pure language-based negotiator to resolve planning conflicts, which has not taken advantage of the multimodal understanding ability of MLLMs. In contrast, previous work V2V-LLM [9] adopts MLLMs to fuse perception features from multiple CAVs and provide answers to perception and planning questions. Although V2V-LLM [9] shows promising results, it has not considered the additional benefit of applying chain-of-thoughts or graph-of-thoughts reasoning on MLLMs.

In this paper, we propose a new graph-of-thoughts reasoning framework for MLLM-based cooperative autonomous

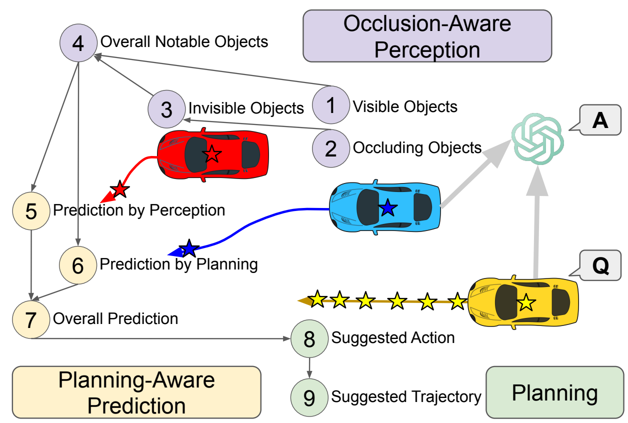


Fig. 1: Illustration of our proposed graph-of-thoughts reasoning framework for cooperative autonomous driving.

driving, as illustrated in Figure 1. In this problem setting, all CAVs share their perception features with a MLLM. Any CAV can ask the MLLM to answer a driving-related question. The MLLM fuses the perception features from all CAVs and performs inference to provide the answer. To verify the effectiveness of our proposed graph-of-thoughts, we first curate the training and testing QA data samples of our proposed V2V-GoT-QA dataset, which is built on top of the V2V4Real [13] cooperative driving dataset. Our V2V-GoT-QA includes 9 types of different QA samples, as illustrated in Figure 2. More specifically, our newly proposed **occlusion-aware perception** questions (Q1 - Q4) consider visible, occluding, and invisible objects. Our newly proposed **planning-aware prediction** questions (Q5 - Q7) include prediction by perception features and prediction by other CAVs' current planned future trajectories. Our planning questions (Q8 - Q9) provide the suggested action settings and suggested waypoints of future trajectories to avoid potential collisions. Different types of QA are connected by a directed edge in our proposed graph-of-thoughts, as shown in Figure 1. The answer of the parent node QA is used as the input context of the child node QA. In addition to the newly curated dataset, we also develop our baseline model V2V-GoT, as shown in Figure 3.

In our experiments, we compare the final planning performance of our proposed V2V-GoT and other baseline methods from the prior work V2V-LLM [9], including *no fusion*, *early fusion*, and *intermediate fusion*. Our experimental results show that our proposed V2V-GoT outperforms all other baselines in the planning tasks, achieving the lowest collision rates and L2 errors between the ground-truth trajectories and model outputs.

Our contribution can be summarized as follows.

- We present a novel graph-of-thoughts reasoning framework for MLLM-based cooperative autonomous driving. The approach includes QA types designed specifically for cooperative driving, including **occlusion-aware perception** and **planning-aware prediction**.
- We curate the V2V-GoT-QA dataset and develop the V2V-GoT model to establish the benchmark and baseline method for future comparative research on MLLM-based cooperative autonomous driving with graph-of-thoughts reasoning.
- Experimental results demonstrate the effectiveness of V2V-GoT in improving overall perception, prediction, and planning performance for cooperative autonomous driving.

II. V2V-GoT-QA Dataset

A. Dataset Details

Following V2V-LLM [9], we use V2V4Real [13] as the base dataset and curate various QA pairs. V2V4Real [13] has 7105 training frames and 1993 testing frames. For each frame that is 3 seconds before the end of each driving sequence in V2V4Real [13], we create one QA pair for each CAV per QA type. Overall, the resulting V2V-GoT-QA dataset has 110610 training and 31014 testing QA pairs.

B. Dataset Curation

We use the ground-truth bounding box annotations, each CAV and object's future trajectory, and their geometric relationships in V2V4Real [13] to curate the QA pairs of V2V-GoT-QA dataset. In our proposed graph-of-thoughts structure, the answer of the parent node QA is used as the input context of the child node QA. For training data, we use the ground-truth answer to generate the input context of the child node QA. During inference, we use the model inference output of the parent node QA to generate the input context of the child node QA.

The QA types included in the V2V-GoT-QA dataset span tasks relating to Perception (Q1 - Q4), Prediction (Q5 - Q7), and Planning (Q8 - Q9). Details of each QA design rationale, how they are connected, dataset curation, and evaluation metrics are described in the following.

C. Perception QAs

The main goal of the perception QAs is to identify notable objects near ego CAV's current planned future trajectory. We propose the idea of **occlusion-aware perception** and divide this task into subtasks of identifying notable objects that are visible (Figure 2a) and invisible (Figure 2c) to the ego CAV separately and then merge the results. This divide-and-conquer design can potentially make it easier for the model to learn which part of the perception feature maps to focus on in order to find the notable objects.

Q1. Visible Notable Objects (Figure 2a): We ask the MLLM to identify notable objects that are visible to the ego CAV and near its current planned future trajectory. Six waypoints from a planned future trajectory in the next 3 seconds are used as the reference trajectory in the question. To generate the ground-truth answer in the training data, at most 3 ground-truth objects within 10 meters of the reference trajectory that are detected by the ego CAV's individual 3D object detector are used as the ground-truth answer. We use the F1 score as the evaluation metric.

Q2. Occluding Objects (Figure 2b): Before identifying notable objects invisible to the ego CAV, locating the occluding objects first may provide useful information. To generate the ground-truth answer in the training data, at most 3 ground-truth objects closest to the ego CAV that are not occluded by other objects are used as the answer. We use the F1 score as the evaluation metric.

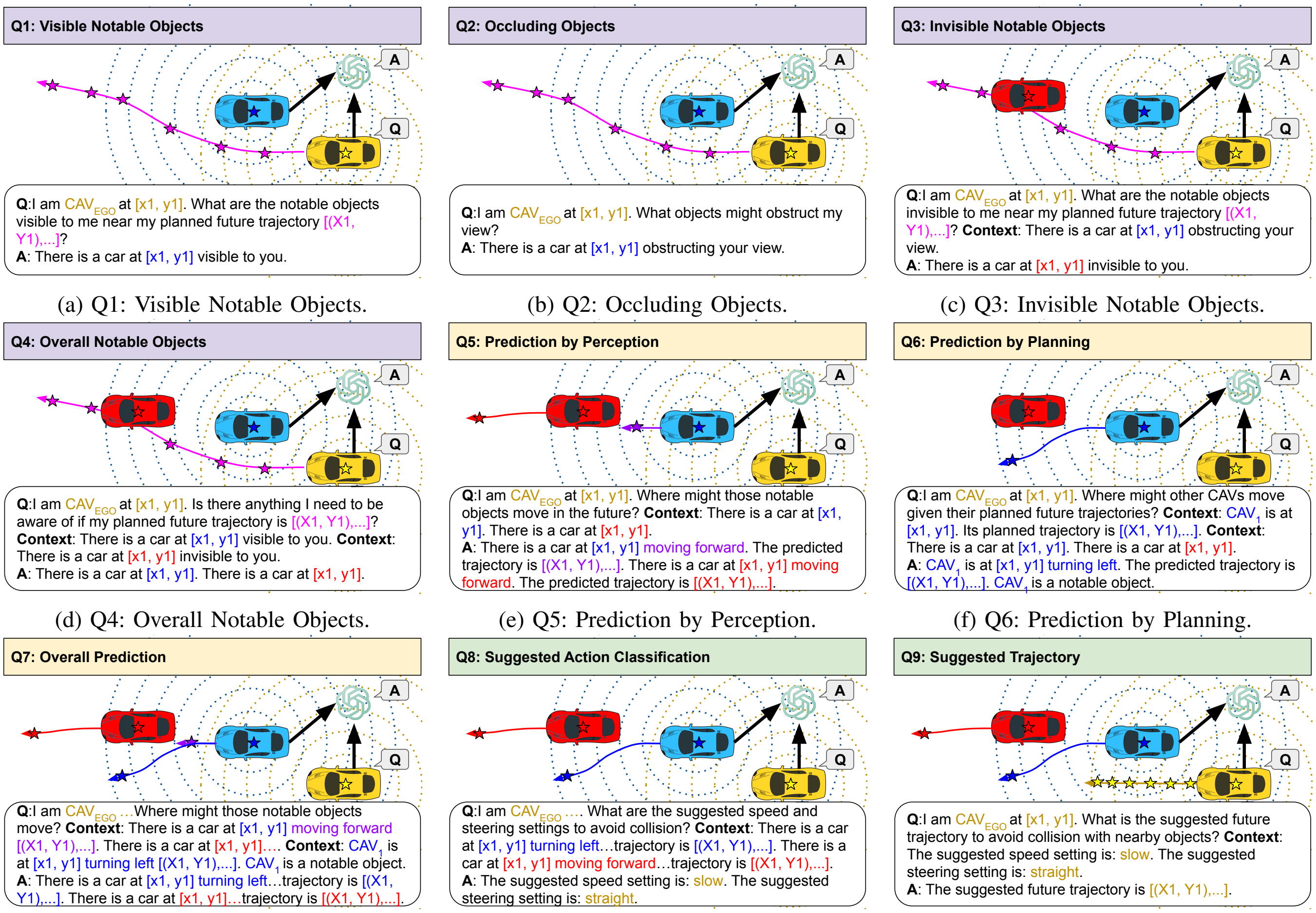
Q3. Invisible Notable Objects (Figure 2c): We ask the MLLM to identify notable objects that are invisible to the ego CAV and near its current planned future trajectory. In addition to the question, we also provide the input context from the answer of Q2. Occluding Objects (Figure 2b). Such context could be useful for the model to identify notable objects that are occluded from the ego CAV's field of view by focusing on the occluded region but in other CAVs' perception feature maps. To generate the ground-truth answer in the training data, at most 3 ground-truth objects within 10 meters of the reference trajectory that are not detected by the ego CAV's individual 3D object detector are used as the answer. We use the F1 score as the evaluation metric.

Q4. Overall Notable Objects (Figure 2d): This question simply takes the answers from Q1. Visible Notable Objects (Figure 2a) and Q3. Invisible Notable Objects (Figure 2c) as the input context and merges them to generate the final notable object identification answer, which are then used as the input context for the subsequent prediction questions. We use the F1 score as the evaluation metric.

D. Prediction QAs

Predicting the future trajectories of notable objects is critical for the final planning task to suggest a future trajectory for the ego CAV that can avoid potential collisions. We propose the idea of **planning-aware prediction** to the graph-of-thoughts reasoning and divide the prediction task into two parts: prediction by perception (Figure 2e) and prediction by planning (Figure 2f), and then merge their results to generate the final prediction output.

Q5. Prediction by Perception (Figure 2e): We ask the MLLM to predict the future trajectories of the notable objects and classify their movement into one of the following 4 categories: *moving forward*, *turning left*, *turning right*, and *staying at the same location*. In addition to the question, the MLLM also takes the answer of Q4. Overall Notable Objects (Figure 2d) as context. With this context, the MLLM knows where the notable objects are at the current timestep. This prediction process relies mainly on the perception features in the current and previous timesteps from all CAVs. We use the ground-truth future trajectories of the notable objects and perform motion classification with heuristic threshold values



(a) Q1: Visible Notable Objects. (b) Q2: Occluding Objects. (c) Q3: Invisible Notable Objects.

(d) Q4: Overall Notable Objects. (e) Q5: Prediction by Perception. (f) Q6: Prediction by Planning.

(g) Q7: Overall Prediction. (h) Q8: Suggested Action Classification. (i) Q9: Suggested Trajectory.

Fig. 2: Illustration of V2V-GoT-QA's 9 types of QA pairs: Perception (Q1 - Q4), Prediction (Q5 - Q7), and Planning (Q8 - Q9). The black arrows pointing at the MLLM indicate the perception data from CAVs. Other colored arrows represent predicted or suggested future trajectories.

on the trajectories as the ground-truth answers. We use L2 error as the evaluation metric.

Q6. Prediction by Planning (Figure 2f): If a notable object suddenly accelerates, decelerates, or changes directions, it may be difficult to predict its future trajectory by only observing its current and past locations and motions. However, if a notable object is also a CAV, it can directly share its planned future trajectory with other CAVs. That planning information could be used as a more accurate predicted future trajectory to help other CAVs' prediction and planning procedures. In Q6, we include the current planned future trajectories of other CAVs and the answer of Q4. Overall Notable Objects (Figure 2d) as the input context in the question. To generate the answer, we perform motion classification in the same manner as for Q5. We further identify whether other CAVs are notable objects or not in Q6's answer. To measure performance, we calculate the binary classification accuracy in determining whether the model can correctly identify if other CAVs are notable objects or not.

Q7. Overall Prediction (Figure 2g): This question takes the answers from Q5. Prediction by Perception (Figure 2e) and Q6. Prediction by Planning (Figure 2f) as the input context and merges them to generate the final prediction answer. If the answer of Q6 indicates that another CAV is a notable object, the model is expected to learn to use that CAV's planned future trajectory as its predicted future trajectory. The merged prediction output is then used as the input context for subsequent planning tasks. To measure performance, we compute L2 errors on the prediction output.

E. Planning QAs

We first classify the suggested action's speed and steering settings, and then use them as the input context to provide the suggested future trajectory.

Q8. Suggested Action Classification (Figure 2h): We use the prediction results from the answer of Q7. Overall Prediction (Figure 2g) as the input context and ask the MLLM to provide the suggested speed and steering settings. We adopt a similar approach from DriveLM [6]. The speed setting can be one of the following 5 categories: *fast*, *moderate*, *slow*, *very slow*, and *stop*. The steering setting can be one of the following 5 categories: *left*, *slightly left*, *straight*, *slightly right*, and *right*. We classify the speed and steering settings by applying heuristic threshold values on the average difference between consecutive waypoint coordinates. To

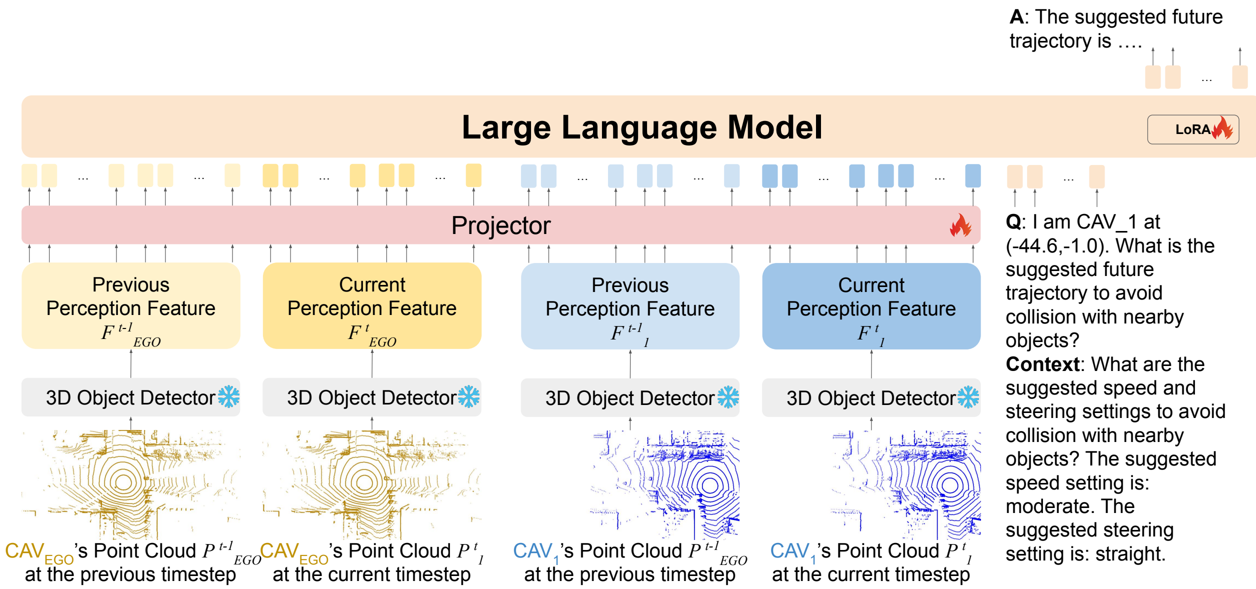


Fig. 3: Model architecture of V2V-GoT.

measure performance, we calculate the L1 errors between the output and ground-truth answer indices on the suggested speed and steering settings. For example, if the output speed and steering settings of a sample are *fast* and *slightly left*, while the ground-truth answer settings are *very slow* and *right*, the L1 error of this sample is $|0-3|+|1-4|=6$.

Q9. Suggested Trajectory (Figure 2i): We use the suggested speed and steering setting from the answer of Q8. Suggested Action Classification (Figure 2h) as context and ask the LLM to provide the 6 waypoints of the suggested future trajectory for the next 3 seconds that avoids collisions. We use the ground-truth future trajectory as the ground-truth answer. To measure performance, we calculate the L2 errors and collision rates.

III. V2V-GoT Model

We extend V2V-LLM [9]'s model to build our V2V-GoT model, as shown in Figure 3. LLaVA [14] is used as the base MLLM architecture. Unlike the original LLaVA [14] that uses an image encoder, we apply a LiDAR-based 3D object detector, PointPillars [15], to extract perception features from each individual CAV's point cloud. Unlike V2V-LLM [9]'s model that only uses the perception features at the current timestep, our model uses the perception features at the current and previous timesteps from all CAVs as the input of the project layers in the MLLM to generate the visual tokens. The MLLM takes the visual tokens and the language tokens from the question and the context as input and generates the final answer in the natural language format. During training, we only train the projector layers and the LoRA parts of the model and freeze the remaining parts.

IV. Experiment

A. Baseline Methods

We adopt V2V-LLM [9] and its baseline methods with different fusion approaches, such as *no fusion*, *early fusion*, and *intermediate fusion*, to compare with our proposed V2V-GoT. In *no fusion*, the MLLM only takes a single CAV's perception features as visual input In *early fusion*, the union of the point cloud of both CAVs is used to extract perception features. In *intermediate fusion*, cooperative 3D object detectors are used to extract perception features as MLLM's visual input.

B. Quantitative Results

Table I shows the testing performance of V2V-GoT in the planning task of V2V-GoT-QA in comparison with baseline methods. Our newly proposed V2V-GoT is seen to achieve the best final planning performance with the lowest L2 errors and collision rates compared to all baselines with different fusion approaches. In particular, the results indicate that the introduction of graph-of-thoughts reasoning designed for cooperative autonomous driving indeed boosts the performance in cooperative autonomous driving planning tasks.

C. Ablation Study

To further verify the impacts of our newly proposed ideas of **occlusion-aware perception** and **planning-aware prediction** in our graph-of-thoughts reasoning framework, we perform an ablation study on two additional graph structures: a *simplified perception graph* and a *simplified prediction graph* The ablation results are given in Table II.

1) Simplified Perception Graph: To verify the impact of **occlusion-aware perception**, we create the *simplified perception graph* reasoning procedure, where we directly ask the model to locate all notable objects without considering Q1, Q2, or Q3 first. We create additional training data samples of Q4. Overall Notable Objects (Figure 2d) but without context from the answer of Q2 and Q3. Then we train the model with newly generated data samples and the existing training dataset, and perform testing inference directly starting from Q4. Overall Notable Objects (Figure 2d). From Table II, we can see that the *simplified perception graph* results in worse perception performance in Q4 and

TABLE I: Testing performance of V2V-GoT in the planning task of V2V-GoT-QA dataset, in comparison with baseline methods, which are adopted from V2V-LLM [9]. To have a fair comparison to our proposed V2V-GoT, we modified baseline methods to also take perception features at the current and the previous timesteps as visual input. L2: L2 distance error. CR: Collision rate. Comm: Communication cost. In each column, the **best** results are in boldface, and the second-best results are in underline.

Method	L2 (m) ↓				CR (%) ↓				Comm(MB) ↓
	1s	2s	3s	Average	1s	2s	3s	Average	
No Fusion	3.47	5.79	8.26	5.84	1.48	4.24	7.72	4.48	**0**
Early Fusion	3.48	5.61	7.82	5.63	1.16	3.51	5.66	3.44	1.9208
Intermediate Fusion									
AttFuse [1]	3.65	6.21	8.75	6.20	1.19	4.41	6.38	3.99	0.4008
V2X-ViT [2]	3.46	5.80	8.19	5.81	1.45	4.24	6.59	4.09	0.4008
CoBEVT [3]	3.38	5.42	7.46	5.42	1.31	4.41	5.75	3.82	0.4008
LLM Fusion									
V2V-LLM [9]	2.90	4.91	6.98	4.93	0.75	2.87	4.93	2.85	0.4068
V2V-GoT (Ours)	**1.65**	**2.63**	**3.59**	**2.62**	**0.12**	**1.92**	**3.45**	**1.83**	0.4068

TABLE II: Testing performance of V2V-GoT in all question-answering tasks of V2V-GoT-QA dataset, in comparison with different inference graphs. F1: F1 score. L2: L2 distance error. L1: L1 distance error. CR: Collision rate. Comm: Communication cost. In each column, the **best** results are in boldface.

Method	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9		Comm(MB) ↓
	F1 ↑	F1 ↑	F1 ↑	F1 ↑	L2 (m) ↓	Accuracy	L2 (m) ↓	L1 ↓	L2 (m) ↓	CR (%) ↓	
Simplified Perception	-	-	-	58.0	8.18	82.0	7.66	0.0958	2.84	1.89	0.4068
Simplified Prediction	52.5	30.1	44.0	**60.8**	**8.05**	-	-	0.0900	3.24	2.17	0.4068
V2V-GoT (Ours)	52.5	30.1	44.0	**60.8**	**8.05**	**87.4**	**7.61**	**0.0876**	**2.62**	**1.83**	0.4068

worse subsequent prediction and planning performance in Q7 and Q9, compared to our proposed V2V-GoT. This result indicates the importance of our occlusion-aware perception design in our final proposed graph-of-thoughts.

2) Simplified Prediction Graph: To verify the impact of **planning-aware prediction**, we create the *simplified prediction graph* reasoning procedure, where we exclude Q6. Prediction by Planning (Figure 2f) and Q7. Overall Prediction (Figure 2g) from the graph-of-thoughts. We directly use the answer of Q5. Prediction by Perception (Figure 2e) and the final prediction result during the inference. From Table II, we can see that this *simplified prediction graph* approach also results in poorer planning performance than our proposed V2V-GoT, due to the poorer prediction result, indicating the importance of our planning-aware prediction design.

V. Qualitative Results

Figures 4, 6, and 5 show the qualitative testing results of our proposed V2V-GoT in Q4. Overall Notable Objects (Figure 2d), Q7. Overall Prediction (Figure 2g), and Q9. Suggested Trajectory (Figure 2i), respectively. The input context information of each question comes from the model inference output of the associated parent question(s). In general, we can observe that our method generates perception, prediction, and planning output results close to ground-truth answers. More qualitative results can be found in the accompanying video.

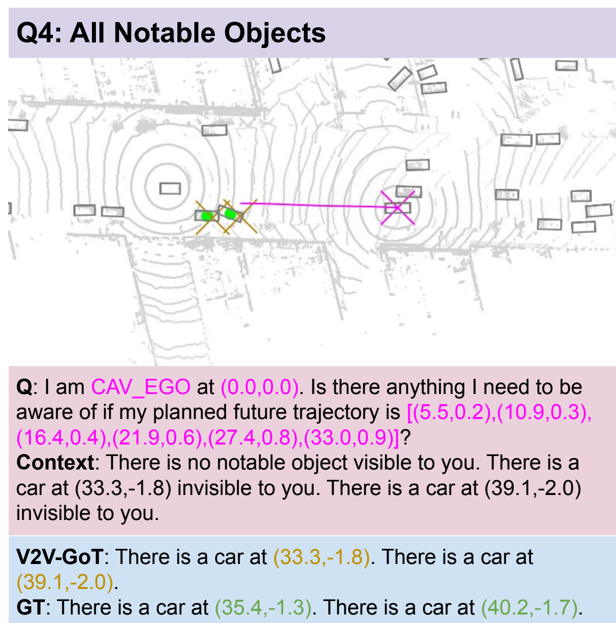
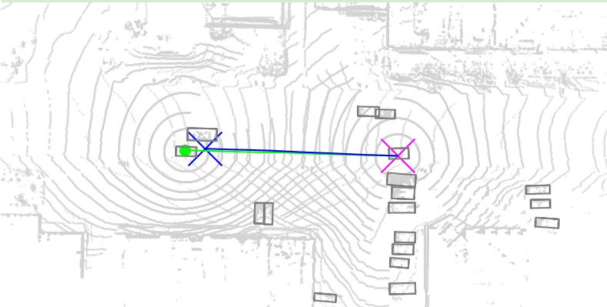


Fig. 4: Qualitative testing sample result of V2V-GoT on Q4. Overall Notable Objects (Figure 2d). The context information is from the testing inference output of parent question Q3. Invisible Notable Objects (Figure 2c) and Q1. Visible Notable Objects (Figure 2a). Magenta ×: current location of the asking CAV. Magenta curve: reference trajectory in the question. Yellow ×: model output. Green ○: ground-truth answer.

VI. Conclusion

In this work, we propose a novel graph-of-thoughts reasoning framework for MLLM-based cooperative autonomous driving. Our proposed graph-of-thoughts includes novel reasoning steps, such as occlusion-aware perception and planning-aware prediction, which are designed to take advantage of V2V information sharing in cooperative driving scenarios and the multimodal understanding ability of MLLMs. To verify the effectiveness of our proposed ideas, we curate the V2V-GoT-QA dataset and develop the V2V-GoT model.

Q9: Suggested Trajectory



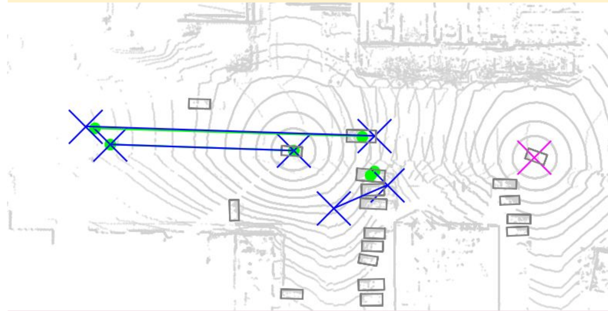
Q: I am CAV_1 at (-44.6,-1.0). What is the suggested future trajectory to avoid collision with nearby objects?
Context: What are the suggested speed and steering settings to avoid collision with nearby objects? The suggested speed setting is: moderate. The suggested steering setting is: straight.

V2V-GoT: The suggested future trajectory is [(-37.9,-0.8),(-31.3,-0.5),(-24.0,-0.2),(-17.8,0.1),(-10.8,0.3),(-3.9,0.5)].
GT: The suggested future trajectory is [(-36.4,-0.8),(-28.3,-0.6),(-20.2,-0.3),(-13.1,-0.2),(-5.8,0.0),(0.3,0.2)].

Fig. 5: Qualitative testing sample result of V2V-GoT on Q9. Suggested Trajectory (Figure 2i). The context information is from the testing inference output of parent question Q8. Suggested Action Classification (Figure 2h). Magenta ×: current location of the asking CAV. Blue curve and ×: model output of the suggested future trajectory and the ending waypoint. Green curve and ○: ground-truth answer.

Our experimental results show that our proposed method outperforms all other baseline methods. Moreover, the ablation study further indicates that our innovative designs in the graph-of-thoughts, such as occlusion-aware perception and planning-ware prediction, improve overall cooperative perception, prediction, and planning performance.

Q7: Overall Prediction



Q: I am CAV_1 at (-50.9,-1.4). Where might those notable objects move in the future?
Context: There is a car at (0.0,0.0) moving forward. The predicted future trajectory is [(50.5,1.9)]. There is a car at (-17.0,3.1) moving forward. The predicted future trajectory is [(43.9,5.1)]. There is a car at (-19.8,-7.3) moving forward. The predicted future trajectory is [(-8.5,-12.2)]. There is a car at (-49.2,-1.3) moving forward. The predicted future trajectory CAV_EGO is at (0.0,0.0) moving forward. Its planned future trajectory is [(38.8,1.3)]. CAV_EGO is a notable object.

V2V-GoT: There is a car at (0.0,0.0) moving forward. The predicted future trajectory is [(38.8,1.3)]. There is a car at (-17.0,3.1) moving forward. The predicted future trajectory is [(43.9,5.1)]. There is a car at (-19.8,-7.3) moving forward. The predicted future trajectory is [(-8.5,-12.2)].
GT: There is a car at (0.0,0.0) moving forward. The predicted future trajectory is [(38.8,1.3)]. There is a car at (-14.4,3.0) moving forward. The predicted future trajectory is [(42.1,4.7)]. There is a car at (-16.3,-5.2) moving forward. The predicted future trajectory is [(-17.1,-4.3)].

Fig. 6: Qualitative testing sample result of V2V-GoT on Q7. Overall Prediction (Figure 2g). The context information is from the testing inference output of parent question Q5. Prediction by Perception (Figure 2e) and Q6. Prediction by Planning (Figure 2f). Magenta ×: current location of the asking CAV. Blue line and ×: model output of the predicted future trajectories, starting, and ending waypoints of notable objects. Green line and ○: ground-truth answer.

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